Code-Layout, Readability and Reusability

Date	15 July 2026
Team ID	SWTID-2026-7100
Project Name	Credit card Approval Prediction
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	YES	-
2	Proper File Structure	Files and folders are logically organized	YES	-
3	Meaningful Variable Name	Variables reflect their purpose clearly	YES	-
4	Function / Method Names	Functions are descriptively named	YES	-
5	Code Comments	Inline and block comments explain logic	YES	-
6	Modular Design	Code is split into reusable functions/modules	YES	-
7	No Redundant Code	Duplicate or unused code is removed	YES	-
8	Error Handling	Exceptions and errors are handled gracefully	YES	-

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level
1	Data Preprocessing Module (`data_cleaning()`)	Python, Pandas	Used during model training and can be reused for new datasets.	High
2	Random Forest Prediction Model	Python, Scikit-learn	Reused in the Flask application for real-time predictions.	High

3	Flask Backend (`app.py`)	Python, Flask	Handles routing and prediction requests; reusable for future ML projects.	High
4	HTML Input Form (`index.html`)	HTML5, Bootstrap, JavaScript	Can be reused for other prediction-based web applications.	Medium
5	Result Page (`result.html`)	HTML5, Bootstrap	Reusable for displaying prediction or classification results in similar applications.	Medium
6	CSS Stylesheet (`style.css`)	CSS3	Reusable across multiple Flask web applications.	High
7	Model Loading Module (`joblib.load()`)	Python, Joblib	Can be reused to load any trained machine learning model.	High
8	Feature Encoding Logic	Python, Scikit-learn (LabelEncoder)	Reusable for preprocessing categorical features in ML projects.	Medium

Overall Code Quality Assessment:

Aspect	**Rating (1–5)**	**Comments**
Code Layout & Structure	**5**	Well-organized project structure with separate folders for templates, static files, and backend code.
Readability	**5**	Meaningful variable names, proper indentation, and clear coding style make the code easy to understand.
Reusability	**5**	Data preprocessing, model loading, prediction, and Flask components are modular and reusable in similar ML projects.
Documentation / Comments	**4**	Sufficient comments explaining preprocessing, model training, prediction flow, and Flask routes. Additional documentation can further improve maintainability.
Overall Score	**4.75 / 5**	The project follows good coding practices, has a modular design, and is easy to maintain and extend.